Chapter 11: Dual Nature of Radiation and Matter

Formula Sheet + Tips & Tricks

Formula Sheet

Constants used in this chapter:
$h = 6.6 \times 10^{-34}$ J s $\quad c = 3 \times 10^{8}$ m/s $\quad e = 1.6 \times 10^{-19}$ C
$m_e = 9.1 \times 10^{-31}$ kg $\quad m_p = m_n = 1.67 \times 10^{-27}$ kg $\quad 1$ Å$= 10^{-10}$ m

I. Photon Energy, Momentum, Work Function

Photon Energy

$$E = h\nu = \frac{hc}{\lambda}$$

ν = frequency, λ = wavelength

Photon Momentum

$$p = \frac{E}{c} = \frac{h}{\lambda}$$

momentum carried by a photon

Dynamic Mass of Photon

$$m = \frac{E}{c^2}$$

rest mass of a photon is always zero

Threshold Frequency

$$\nu_0 = \frac{W_0}{h}$$

W_0 = work function of the metal

Threshold Wavelength

$$\lambda_0 = \frac{hc}{W_0}$$

longest wavelength that can cause emission

Photon Count from Power

$$n = \frac{P}{E_{\text{photon}}}$$

P = power of source, n = photons/s

II. Einstein's Photoelectric Equation

Einstein's Equation

$$E_{k(\max)} = h\nu - W_0 = \frac{hc}{\lambda} - W_0$$

$E_{k(\max)}$ = max. KE of ejected electron

Max. Speed of Photoelectron

$$v_{\max} = \sqrt{\frac{2E_{k(\max)}}{m_e}}$$

from max. kinetic energy

III. Stopping Potential

Stopping Potential Relation

$$eV_0 = h\nu - W_0$$

V_0 = stopping potential (volt)

Slope of V_0–ν Graph

$$\text{slope} = \frac{h}{e}$$

gives Planck's constant from graph

Intercepts of V_0–ν Graph

$$\nu_0 = \frac{W_0}{h}, \quad V_0\Big|_{\nu=0} = -\frac{W_0}{e}$$

ν_0 = x-intercept, $-W_0/e$ = y-intercept

IV, V & VI. de Broglie Wavelength

de Broglie Wavelength (general)

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

p = momentum of the particle

de Broglie Wavelength (KE given)

$$\lambda = \frac{h}{\sqrt{2mE_k}}$$

E_k = kinetic energy of the particle

de Broglie Wavelength (electron, accelerating p.d. V)

$$\lambda = \frac{h}{\sqrt{2m_e eV}} \approx \frac{12.27}{\sqrt{V}}\ \text{Å}$$

V in volts, standard result for electrons

Tips, Tricks & Hacks

- Work function $\phi_0 = h\nu_0$ sets the threshold frequency; below ν_0, no photoelectrons are emitted *no matter how intense* the light — intensity controls number of photoelectrons, frequency controls their maximum kinetic energy, never mix the two roles up.
- Einstein's photoelectric equation: $KE_{max} = h\nu - \phi_0 = eV_0$ — plotting V_0 vs ν gives a straight line of slope h/e and intercept related to ϕ_0; this graph is a favourite numerical/derivation.
- Stopping potential depends only on frequency of incident light, not intensity — increasing intensity increases photocurrent (saturation current) but leaves V_0 unchanged at fixed frequency.
- de Broglie wavelength $\lambda = \frac{h}{p}$ applies to *all* matter, not just electrons — for a particle accelerated through potential V, use $\lambda = \frac{h}{\sqrt{2mqV}}$ directly instead of finding v first, it saves a step.
- For electrons specifically, the handy shortcut is $\lambda = \frac{1.227}{\sqrt{V}}$ nm (V in volts) — worth memorising to skip algebra in MCQs.

- Heavier particles (protons, alpha particles) have much shorter de Broglie wavelength than electrons for the same kinetic energy or same accelerating voltage, since $\lambda \propto \frac{1}{\sqrt{m}}$ — a quick comparison trick for ranking questions.
- Davisson–Germer experiment is the experimental proof of de Broglie's hypothesis (electron diffraction) — commonly asked as a one-line "which experiment confirms..." question.